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The role of personality traits in athlete selection: A systematic review

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ABSTRACT

Background: Personality traits are increasingly used in athlete selection, yet their predictive validity varies across sports, levels, and cultural contexts. This review identifies which traits best predict performance and through what mechanisms.

Methods: Following PRISMA and a PROSPERO-registered protocol, we searched five databases through March 2025. Twenty-two cross-sectional studies ($N = 4181$ athletes; mean age 15–37 years; 14 countries; 18 sports) met inclusion criteria. Study quality was assessed via the STROBE checklist.

Results: Higher conscientiousness and extraversion consistently predicted superior performance (effect sizes ~0.26–0.70), while neuroticism was negatively related (β up to −0.25). Three mechanisms emerged: (1) direct neurobiological effects via dopaminergic sensitivity and prefrontal regulation; (2) indirect psychological mediation through motivation and self-regulation; and (3) contextual moderation by sport type, competitive level, and cultural background.

Conclusions: Conscientiousness, extraversion, and low neuroticism are robust, dynamic predictors of athletic success. Incorporating multi-trait assessments tailored to sport and culture can enhance talent identification. Future longitudinal, multi-method research with standardized metrics is warranted.

1. Introduction

Athlete selection serves as a fundamental process in identifying and cultivating sporting talent, particularly at the elite level, where efficiently pinpointing future champions from a broad talent pool presents both strategic significance and methodological complexity, as emphasized by Williams and Reilly (2000). Globally, it is estimated that over 60 million individuals are actively engaged in competitive sports across various age groups and disciplines. In this context, efficiently and accurately identifying individuals with genuine long-term potential has become a question of critical practical and scientific relevance.

Since the institutionalization of modern competitive sports in the 20th century, structured athlete selection systems have been widely established across countries and organizational levels. For example, the United States primarily relies on the collegiate athletic system (NCAA) and multi-stage professional league scouting structures, while China has developed a state-led, tiered youth selection system encompassing schools, provincial teams, and national training bases. These selection models often depend on short-term physical or technical indicators which, although easy to quantify (e.g., sprint speed, anthropometry, basic skills), tend to overlook latent characteristics such as psychological adaptability, motivation, and stress tolerance—key factors for sustained high-level performance, as noted by Vaeyens et al. (2008) and Gledhill et al. (2017).The absence of psychological evaluation creates a critical gap in assessing athletes' adaptive capacities, thereby weakening the predictive power of purely physical metrics. With the gradual integration of psychological assessments—particularly personality trait evaluations—selection systems have begun to bridge this gap and evolve into more comprehensive, long-term development-oriented frameworks. In recent years, national institutions such as UK Sport and the General Administration of Sport of China have progressively incorporated psychological assessments into early selection phases to enhance the accuracy and effectiveness of talent identification. This evidence highlights the critical need to incorporate psychological profiling, particularly personality assessment, as an integral component of modern selection systems. Research has consistently demonstrated that psychological factors contribute substantially to the accurate identification of athletes with genuine potential for elite performance, as shown by Allen et al. (2013) and further supported by Piepiora (2021). The integration of these multidimensional evaluation frameworks represents a

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significant advancement in talent identification methodologies.

Personality traits, as defined by Allen and Laborde (2014), are enduring patterns of emotion, cognition, and behavior that remain relatively stable over time and across situations, making them highly predictive in long-term talent identification systems. Compared to state-like psychological variables that are more susceptible to situational fluctuations (e.g., mood, fatigue), personality traits provide a more consistent psychological basis for understanding athletes' behavioral tendencies during training, competition, and career development. Currently, athlete personality assessment primarily relies on two categories of measurement tools: general structural personality inventories and sport-specific psychological trait scales. Among structural models, the Five-Factor Model of Personality (FFM)—which includes openness, conscientiousness, extraversion, agreeableness, and neuroticism—has become the most widely used framework, as noted by McCrae and Allik (2002), due to its robust psychometric properties and cross-cultural applicability. Frequently adopted FFM-based instruments include the NEO Personality Inventory and the Big Five Inventory. In contrast, sport-specific trait scales focus more closely on psychological resources critical to athletic performance, such as the Psychological Characteristics of Developing Excellence Questionnaire (PCDEQ) and the Athletic Coping Skills Inventory (ACSI), which assess attributes like self-regulation, stress tolerance, and competitive motivation. These tools are generally used to complement, rather than replace, general personality inventories.

In this review, athletic performance is operationalized based on two major categories of measurement: objective indicators and subjective evaluations. Objective indicators include quantifiable metrics such as competition rankings, win–loss records, match statistics (e.g., goals, assists, accuracy), and medal attainment. Subjective evaluations encompass expert assessments, including coach ratings, selection status (e.g., elite vs. non-elite classification), and self-reported performance appraisals. This multidimensional operationalization allows for comprehensive synthesis across studies using varied performance criteria.

2. Associations between personality traits and athletic performance

Personality traits have been shown to effectively differentiate athletes at different levels of competition and have emerged as an important area of research in sport psychology. Recent meta-analyses and longitudinal studies consistently identify several core trait characteristics in elite athletes. Among these, higher levels of extraversion and conscientiousness are frequently observed, as evidenced by Li et al. (2022) in their findings on elevated social initiative, goal orientation, and disciplined performance. A meta-analysis by Allen et al. (2013) revealed that elite athletes tend to score higher on conscientiousness and agreeableness, with medium-to-large effect sizes. These traits commonly manifest in greater training adherence and strategic goal pursuit (indicative of conscientiousness), as well as improved team cohesion and cooperation (indicative of agreeableness), all of which contribute to enhanced performance in structured competitive environments.

Elite athletes also display greater emotional stability (i.e., lower levels of neuroticism), which supports cognitive clarity under pressure. For instance, Fabbricatore et al. (2023) found that top-level swimmers scored significantly lower in neuroticism. Structural equation modeling indicated that emotional stability exerted an indirect effect on performance by enhancing psychological skills such as positive self-talk, arousal control, and reduced anxiety. Furthermore, self-regulation appears to function as a key intermediary between personality and performance. Roberts et al. (2012) found that conscientiousness moderately predicted self-regulatory abilities, with athletes high in this trait demonstrating stronger goal management and emotional control, ultimately supporting training persistence and long-term success. Trait-driven self-regulation thus emerges as a pivotal contributor to elite athletic development.

It is important to note that the optimal personality profile may vary across sport disciplines. For example, Shuai et al. (2023) observed that elite endurance athletes exhibited relatively lower levels of extraversion and emotional stability, suggesting that personality-performance associations are context-dependent. Accordingly, athlete selection systems should integrate broad trait screening (e.g., conscientiousness, emotional stability), sport-specific alignment (e.g., extraversion and agreeableness in team sports), and developmental sensitivity (e.g., accounting for neuroticism fluctuations in youth athletes) into a comprehensive, context-sensitive psychological assessment framework.

3. Personality and sport preference

Substantial empirical evidence suggests that athletes from different sport types display distinct personality trait profiles. For instance, Shuai et al. (2023) demonstrated that individuals engaged in team sports tend to score higher on extraversion, sociability, and agreeableness, whereas those involved in individual sports typically exhibit greater independence, emotional stability, and cognitive flexibility. To account for such consistent differences, three interrelated theoretical models have been proposed—the Attraction Hypothesis, the Developmental Hypothesis, and the Interactionist Model—which represent progressively integrative perspectives on how personality traits and sport participation influence each other. The Attraction Hypothesis posits that individuals possess relatively stable dispositional traits prior to sport engagement and are naturally drawn to sport types that align with their personalities, as originally proposed by Eysenck (2013). For example, Yang et al. (2024) conducted a meta-analysis indicating a positive association between extraversion and sport participation, with relatively stronger effects observed in team sports. According to conventional criteria, the observed effect size fell within the small-to-moderate range. This suggests that extraverted individuals, who are more sensitive to social rewards and interpersonal stimulation, may be more inclined to select cooperative and interactive sports such as football or basketball.

Conversely, the Developmental Hypothesis suggests that long-term sport participation can shape and reinforce personality traits through behavioral feedback and contextual reinforcement. For instance, Stephan et al. (2018) provide the strongest longitudinal evidence to date for the Developmental Hypothesis. In three large cohort samples (Wisconsin Longitudinal Study Graduates, $N \approx 10{,}000$; Siblings, $N \approx 8000$; MIDUS, $N \approx 5000$; combined $N \approx 23{,}000$), baseline physical inactivity predicted significantly steeper declines in conscientiousness over a 20-year follow-up (standardized $\beta = -0.06$, $p < 0.001$) and smaller but reliable declines in openness ($\beta = -0.05$), extraversion ($\beta = -0.03$), and agreeableness ($\beta = -0.03$) in latent growth models; neuroticism trajectories were unaffected. These results demonstrate that sustained sport participation—or, conversely, inactivity—exerts cumulative behavioral feedback and contextual reinforcement on adaptive traits, thereby stabilizing or enhancing conscientiousness, social engagement, and cooperativeness over time. Such causal evidence directly supports the notion that long-term engagement in physical activity can shape and reinforce personality development. Furthermore, conscientiousness has been identified as a stable predictor of physical activity initiation and maintenance across multiple samples, implying that sustained participation may foster self-regulatory traits like diligence and consistency, as demonstrated by Wilson and Dishman (2015). While these two hypotheses offer valuable but unidirectional explanations, the Interactionist Model provides a more dynamic view, proposing that personality traits and sport contexts mutually shape one another. Initial personality predispositions may influence sport choices, while sport-specific environments can activate and consolidate certain traits over time. Roberts et al. (2005) emphasized that specific sport environments—especially those with high cognitive or emotional demands—can trigger latent personality dimensions, which in turn affect athletic behavior and development. For example, athletes in cognitively demanding sports may

exhibit stronger attentional control and emotional stability due to ongoing trait–task alignment.Taken together, these models illustrate that personality–sport relationships are not static but dynamically regulated through developmental timing, sport-specific demands, and individual dispositions. Understanding these mechanisms can inform the refinement of athlete selection systems by integrating psychological profiling into context-sensitive, developmentally appropriate, and sport-specific talent identification strategies.

4. Mechanisms linking personality traits to athlete selection outcomes

To move beyond traditional correlational research, recent work in sport psychology has proposed three key mechanisms through which personality traits influence athletic selection and performance: direct effects, indirect effects, and moderated effects. Complementing this, research in personality neuroscience shows that stable personality traits are grounded in neurobiological systems that regulate behavior.

At the neurobiological level, approach-related traits (e.g., extraversion and conscientiousness) are linked to dopaminergic activity in regions like the ventral striatum, orbitofrontal cortex (OFC), and medial prefrontal cortex (mPFC), which support reward sensitivity, goal pursuit, and executive control. For example, extraversion has been associated with greater volume and activity in the medial OFC, a region involved in evaluating social rewards.

In contrast, avoidance-related traits (e.g., neuroticism) are associated with the serotonergic system, involving areas such as the amygdala, anterior cingulate cortex (ACC), and hippocampus. These regions regulate emotional responses and stress. High neuroticism corresponds to increased amygdala reactivity and reduced prefrontal control, leading to greater emotional reactivity and anxiety.Conscientiousness—an important predictor of training adherence and self-regulation—has been linked to increased gray matter volume in the lateral prefrontal cortex, particularly the middle frontal gyrus, which supports planning, impulse control, and goal maintenance.

These findings support a biobehavioral framework:

Approach traits (e.g., extraversion, conscientiousness) activate reward-related brain circuits, supporting engagement, persistence, and social interaction.

Avoidance traits (e.g., neuroticism) reflect heightened activity in emotion-processing circuits, increasing stress sensitivity and emotional variability.

These neurobiological patterns help explain how personality traits influence behavior in sport contexts. Importantly, their effects vary by sport demands and environments. This highlights the need for a multilevel model that integrates personality, motivation, and neurobiology to explain athlete selection and performance more fully. Overall, personality traits are increasingly recognized as stable psychological characteristics that relate meaningfully to athletic performance. While evidence is growing, current findings remain fragmented across sport types, competitive levels, and assessment tools. Therefore, this review, following PRISMA guidelines, synthesizes studies on how personality relates to athlete selection, with particular attention to underlying mechanisms and contextual differences. The next section outlines the methodology used for identifying and analyzing relevant studies.

5. Methods

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline, as outlined by Page et al. (2021), which outlines key stages including literature search, study screening, data extraction, bias assessment, and result synthesis. The review protocol was prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO; registration number: CRD420251007053), and the full record is publicly available in the PROSPERO database.

5.1. Search strategy

A systematic search was conducted on 5 March 2025 across five databases: PubMed, Embase, APA PsycINFO®, Scopus, and Web of Science. The search covered all records from database inception to the search date. To ensure reproducibility, a combination of controlled vocabulary (e.g., MeSH, Emtree) and free-text terms was applied consistently across databases, targeting two core concepts: (1) personality traits and (2) athletic performance. Boolean operators (AND, OR) were used to structure the search (see Appendix 1 for the complete search syntax).

5.2. Study selection

Two independent reviewers (R1 and R2) conducted the initial search and removed duplicates using EndNote X9, followed by manual verification to exclude records with identical titles, DOIs, or authorship. From an initial yield of 4550 articles, 2955 unique records were retained for title and abstract screening.During this phase, both reviewers independently evaluated all records based on predefined inclusion criteria. Inter-rater agreement reached 94.7 % (Cohen's $\kappa = 0.87$), indicating strong consistency. A total of 423 potentially eligible studies underwent full-text review. Any disagreements were resolved through discussion or adjudication by a third reviewer (R3). Ultimately, 22 studies fulfilled all eligibility criteria. The complete screening process is illustrated in the PRISMA flow diagram (Fig. 1).

5.2.1. Inclusion criteria

Studies were included if they met all of the following conditions:

1. Population: Involved competitive athletes from any sport discipline (e.g., elite, collegiate, academy-level).
2. Outcomes: Reported on both personality traits (e.g., Big Five, 16PF, or structured self-report questionnaires) and athletic performance indicators. Performance outcomes could be:

   Objective (e.g., competition results, statistics),
   Semi-objective (e.g., coach ratings, team selection),
   Structured self-classifications (e.g., elite vs. non-elite), provided that classification criteria were clearly stated.

3. Design: Employed either quantitative (e.g., observational, experimental) or qualitative (e.g., interview, case-based) designs that reported original data.
4. Data Reporting:

   Quantitative studies must provide extractable results (e.g., means, standard deviations, effect sizes).
   Qualitative studies must present verifiable data excerpts (e.g., thematic codes or participant quotations).

5. Language: Published in English in a peer-reviewed journal.

5.2.2. Exclusion criteria

Studies were excluded if they met any of the following:

1. Publication Type: Review articles, meta-analyses, editorials, conference abstracts, or study protocols without original data.
2. Population: Involved non-athlete participants (e.g., general students without training history) or recreational exercisers without documented competitive involvement or with insufficient training frequency (<4 h/week).
3. Design: Case reports with fewer than five participants, animal studies, or studies lacking comparative or correlational analysis.

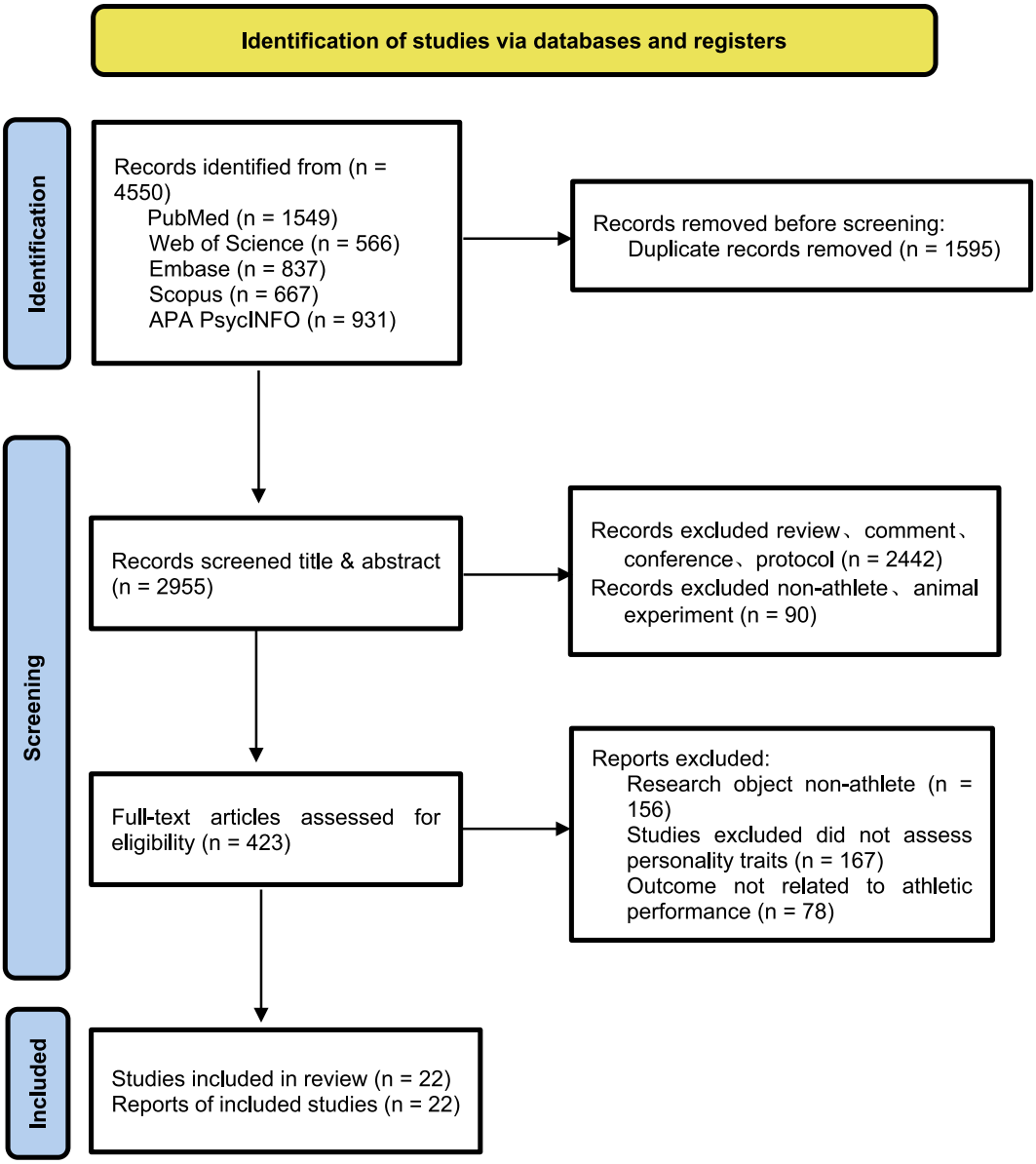

Fig. 1. PRISMA flow diagram of the study selection process.
Note. From 4550 initial records, 1595 duplicates were removed. Of the 2955 titles and abstracts screened, 2532 were excluded. After full-text assessment of 423 articles, 22 studies met the inclusion criteria and were included in this systematic review.

4. Data Availability:

No extractable personality or performance data (e.g., *p*-values only without descriptive statistics),

Use of unstructured or anecdotal personality assessments,

Performance outcomes were unclearly defined or solely based on unclassified self-reports.

Note: Given the methodological heterogeneity across sport psychology research, this review adopted an inclusive but rigorous approach. Studies employing non-commercial yet structured tools or classification frameworks were accepted, provided that measurement rationale and comparative validity were sufficiently detailed.

5.3. Study quality assessment

The quality of the 22 included studies was assessed using the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) checklist, which consists of 22 essential items covering study design, participant selection, data collection, bias control, statistical analysis, and results reporting. Two independent reviewers conducted the quality assessment. For each item, the scoring options were "Yes," "Partly," and "No." Discrepancies in scoring were resolved through discussion or, when necessary, adjudicated by a third senior reviewer.

A summary of the quality assessment results is presented in Table 3. Overall, most studies adequately reported key methodological elements such as study objectives, design, participant characteristics, and analytical strategies. However, several recurrent limitations were identified, including insufficient information on sample size estimation, incomplete reporting on missing data, inadequate description of bias control methods, and limited discussion of study limitations. These issues may contribute to increased risk of bias and should be taken into account when interpreting the findings.

5.4. Data extraction

Data were extracted using a standardized form developed for this review. For each included study, the following information was systematically recorded: author(s), year of publication, country, sample characteristics (e.g., gender distribution, total sample size), type of sport, athletic experience, personality assessment instrument(s), performance evaluation criteria, key findings, and any limitations reported by the authors. These details are presented in Table 1 and Table 2.

Table 1 summarizes the demographic and methodological features of each study, including participant profiles, measurement tools, and study design. Table 2 provides a structured synthesis of the reported associations between personality traits and athletic performance. The findings are organized by trait dimension, performance domain, and effect size indicators (e.g., correlation coefficients, *p*-values, regression weights). Additionally, Table 2 highlights studies that identified moderating or mediating variables, such as genetic markers, cognitive factors, or psychological states (e.g., anxiety).

5.5. Definition of athletic performance

In this review, athletic performance is defined as a multidimensional construct incorporating both objective performance indicators and subjective evaluative assessments that reflect an athlete's competitive effectiveness. Objective metrics include quantifiable competition data such as goals, assists, shooting percentages (e.g., FG%, 3P%), player efficiency indices (e.g., PER, WS), and match-specific statistics. Subjective assessments primarily consist of coach or expert ratings targeting aspects like tactical intelligence, work ethic, coachability, and team contribution. Some studies used composite indices that integrated objective and subjective measures to enhance reliability.In addition, classification-based criteria (e.g., national team selection, NCAA division status, MLB draft level) were adopted as proxies for elite performance. A few studies further applied machine learning methods (e.g., SVM, ANN) to classify athletes by performance tier, with reported accuracies reaching up to 97 %.

5.6. Personality trait assessment tools

In this systematic review, the personality assessment tools used in the included studies were categorized into three main types. The first category comprises general personality inventories, including Five-Factor Model (FFM)-based instruments such as NEO-PI-3, NEO-FFI, BFI, BFI-2, and localized Big Five scales; Cattell's 16 Personality Factor Questionnaire (16PF), widely used for athlete profiling; the Eysenck Personality Questionnaire (EPQ/EPQ-J) and the Freiburg Personality Inventory-Revised (FPI-R), particularly common in endurance sport contexts; as well as tools like the Temperament and Character Inventory (TCI) and the State-Trait Anxiety Inventory (STAI), which focus on emotional and temperamental dimensions. The second category includes sport-specific instruments, such as the Psychological Characteristics of Developing Excellence Questionnaire (PCDEQ), the Athletic Coping Skills Inventory (ACSI), and the Psychological Characteristics Related to Sport Performance Questionnaire (CPRD), which are specifically designed for personality assessment in athletic settings. The third category consists of other or non-standardized tools, including the Bipolar Adjective Scale (BAS), IBM Watson Personality Insights—an AI-driven, black-box tool that analyzes personality from digital content, though flagged for its uncertain psychometric validity—and various other scales measuring motivation, emotion, and identity, such as SMS, SEQ, FNE, MSRS, AIMS, FCZ-KT, and POMS. Whenever possible, only validated instruments were included in this review. Tools with limited or unclear validation (e.g., IBM Watson) were noted accordingly and interpreted with caution.

5.7. Ethical considerations

This study is a systematic review and meta-analysis based entirely on data from previously published literature. No direct experimentation involving human participants was conducted. As such, ethical approval was not required, and the exemption was granted by the Institutional Review Board of [MASKED].

6. Result

6.1. Characteristics of included studies

This systematic review included 22 eligible cross-sectional studies with a total of 4181 participants. The studies were published between 1989 and 2025 and conducted across 14 countries, including the United States, China, Germany, Poland, Australia, Brazil, and several European and Middle Eastern nations. Most samples consisted of adolescents or young adults, with mean ages ranging from 15.31 to 36.78 years. Gender composition varied across studies: three included only female participants, five included only male participants, and fourteen used mixed-gender samples, with the proportion of female participants ranging from 23 % to 70 %.

Regarding study design, 21 studies employed cross-sectional methods, while one utilized a quasi-longitudinal approach. The range of sports investigated was broad. Basketball ($n = 5$), football/soccer ($n = 3$), and mixed-sport samples ($n = 4$) were most common. Other disciplines included handball, endurance events, American football, baseball, taekwondo, BASE jumping, and adaptive sports for athletes with disabilities.

With respect to athletic level, eight studies focused on elite or professional athletes (e.g., national team members, NCAA athletes, or professional league players). The remaining fourteen included amateur, semi-professional, or general competitive-level athletes. This diversity of participant backgrounds enhances the generalizability of the findings across different sport settings.

A wide variety of personality assessment instruments were employed. The majority of studies used tools based on the Five-Factor Model (FFM), such as the NEO Personality Inventory-3 (NEO-PI-3), NEO Five-Factor Inventory (NEO-FFI), Big Five Inventory (BFI, BFI-2), and the International Personality Item Pool–Big Five Markers 20 (IPIP-BFM-20). Additional frameworks included Cattell's 16 Personality Factor Questionnaire (16PF), the Freiburg Personality Inventory-Revised (FPI-R), the Eysenck Personality Questionnaire (EPQ, EPQ-J), and the Temperament and Character Inventory (TCI). Some studies also used sport-specific tools, including the Sport Motivation Scale (SMS) and the Sport Emotion Questionnaire (SEQ), to capture performance-relevant psychological traits.

Athletic performance was assessed using a multidimensional framework. Objective indicators included quantifiable metrics such as competition rankings, match statistics, and medal records. Subjective indicators included coach evaluations and athlete self-reports. Several studies used classification-based performance tiers (e.g., elite vs. non-elite, national team selection), and a few incorporated composite indices or algorithmic models to improve differentiation. This methodological diversity contributed to both the breadth and variability of performance outcomes.

Study quality was assessed using the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) checklist. Eleven studies (e.g., Bonetti et al., 2025; Remiszewska et al., 2020) were rated as high quality, having met more than 85 % of reporting criteria. Nine were rated as moderate quality (65 %–85 % compliance), while two studies (e.g., Nudel et al., 1989; Kinga & Aleksandra, 2024) were rated as low quality due to reporting deficiencies. Despite these variations, most studies identified consistent associations between personality traits—especially conscientiousness and extraversion—and athletic performance. High-quality studies tended to report stronger effect sizes

Table 1
Characteristics of the included studies.

References	Country	Female %	AGE	*N*	Type of sport	Athletic Experience	Personality measurement	Measurement of sports performance	Limitations
Nudel et al. (1989)	USA, Israel	37 %	15.4 (2.4)	*N* = 16 (AG = 16; NAG = 0)	Long Distance Running	Started at 4–12 yrs.; 3–15 yrs. training; 30–105 miles/week; some completed marathon, 30–60 mile races.	Cattell 16 Personality Factor Questionnaire (16PF)	participation types recorded; focus on sport history, physiology, and psychology.	Small sample (*n* = 16); 13 with psychological data; uncontrolled (gender, culture, family); no control for psychological traits; retrospective injury data; some psychological variables unanalyzed.
Garland and Barry (1990)	United States	0 %	NR	*N* = 272 (REG = 101; SUB = 94; SUR = 77)	Football	Scholarship athletes; training years NR; high-level NCAA players	Cattell's Sixteen Personality Factor Questionnaire (16PF)	Grouped by coaches' season summaries into Regulars, Substitutes, and Survivors; analyzed in relation to personality and perceived leadership.	No objective data; cross-sectional; uncontrolled confounders (age, position, training years); males only; limited to southeastern U.S. colleges
Newcombe and Boyle (1995)	Australia	41 % (AG = 38 %; NAG = 48 %)	16.98 (1.02)	*N* = 312 (AG = 212; NAG = 100)	Mixed sports	Athletes with official competition experience at school or club level	① State-Trait Anxiety Inventory (STAI) ② Eysenck Personality Questionnaire (EPQ) ③ Profile of Mood States (POMS)	Grouping: Elite = national/state representative; Non-Elite = general competitive level or participants	Adolescents were sampled; sport experience and position were uncontrolled; POMS may be subject to social desirability bias; cross-sectional design; intragroup differences within sports were not examined.
Egloff and Gruhn (1996)	Germany	32.7 % (AG = 32 %; NAG = 32 %)	30.7 (9.79)	*N* = 159 (AG = 86; NAG = 73)	Endurance sports (long-distance running and triathlon)	Average 6.75 30.7 (NR) hours/ week; ordinary (<4 h/week) vs. outstanding (>11 h/week) groups.	Freiburg Personality Inventory-Revised (FPI-R)	Grouped by endurance athlete status and training volume (ordinary vs. outstanding); training years and motivation assessed via self-report.	Retrospective self-report; no long-term follow-up; no competitive performance data; high-level elite athletes were not included; cross-sectional design.
Piedmont et al. (1999)	United States	100 %	18.5 (NR)	*N* = 68 (AG = 68; NAG = 0)	Women's Soccer	Official collegiate athletes (completed at least one full season)	Bipolar Adjective Scale (BAS)	①Coachability、Athletic ability、Game performance、Team playerness、Work ethic; ② Actual match statistics (goals, assists, appearances, shots); ③ Composite index combining coach ratings and match statistics.	Small sample size (n = 68); missing match data from one team (*n* = 10); cross-sectional design; confounding variables such as player position and experience were not controlled.
Saale-Prasad (2013)	United States	NR	NR	*N* = 176 (AG = 176; NAG = 0)	Mixed sports	NCAA Division I athletes (collegiate competition experience)	NEO Personality Inventory-3 (NEO-PI-3)	Coach-rated overall performance; controlled for athletic ability	Sport type, gender, culture, and training years uncontrolled; no objective performance measures; coach ratings only; cross-sectional design.
Čavala et al. (2013)	Croatia	100 %	15.31 (0.64)	*N* = 70 (AG = 34; NAG = 36)	Handball	Club registered players, specific years not reported	Eysenck Junior Personality Questionnaire (EPQ-J)	National team selection as a high-performance indicator; positions classified as	Small sample size; narrow age range with no long-term follow-up; cross-sectional design; personality

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Table 1 (*continued*)

References	Country	Female %	AGE	*N*	Type of sport	Athletic Experience	Personality measurement	Measurement of sports performance	Limitations
								goalkeeper, back, wing, and center.	assessment limited to three dimensions of the EPQ-J; training years and athletic experience were not controlled.
Ramos-Villagrasa et al. (2013)	Spanish	47.05 %	24.0 (3.77)	*N* = 34 (AG = 34; NAG = 0)	Basketball	Semi-professional players; experience range: 5–28 years.	① Big Five Inventory (BFI-44, Spanish version) ② Sport Motivation Scale (SMS)	① Objective performance: composite performance rating per match (including points, assists, rebounds, turnovers, etc.); ② Performance fluctuations: stability analyzed using the Mean Squared Successive Differences (MSSD) method.	Small sample (*n* = 34); limited to Spanish semi-professional players; no analysis of position, team characteristics, or task difficulty; skill not included; no validation on professional samples; subjective GME parameters.
Geukes et al. (2017)	Germany & Australia	49 %	24.1 (4.33)	*N* = 53 (AG = 53; NAG = 0)	Basketball	10 years of competitive basketball experience; SEABL semi-professional players	① Fear of Negative Evaluation (FNE) ② Movement Specific Reinvestment Scale (MSRS) ③ Athletic Identity Measurement Scale (AIMS)	Free-throw accuracy under low-pressure (30 attempts) and high-pressure (12 official games, 579 data points)	Small sample (*n* = 53); Australian players only; cross-sectional and quasi-longitudinal design; uncontrolled competitive context; pressure manipulation limited to regular-season games
Monasterio et al. (2018)	New Zealand & USA	23 %	36.78 (11.91)	*N* = 83 (AG = 83; NAG = 0)	BASE Jumping	≥10 BASE jumps; mean experience = 6.7 years (max = 28 years)	Temperament and Character Inventory (TCI)	participants were classified as experienced BASE jumpers with ≥10 jumps and average 6.7 years of practice.	Small female sample (*n* = 19); confounders (nationality, culture, training, performance) uncontrolled; cross-sectional; only TCI used without cognitive or emotional measures.
Remiszewska et al. (2020)	Spanish	100 %	18.28 (4.05)	*N* = 134 (AG = 134; NAG = 0)	Female Soccer	N = 134 registered female soccer players (≥3 years competition, >14 yrs); mostly second division, some first division (AG = 134; NAG = 0)	Psychological Characteristics related to Sports Performance Questionnaire (CPRD)	Grouped by competitive level (professional vs. amateur); no specific performance data used; psychological traits analyzed by group.	Convenience sample; cross-sectional design; age and training years not included as covariates; low reliability for some scales (e.g., psychological skills); performance, position, and training load uncontrolled.
Ruiz-Esteban et al. (2020)	Poland	NR	17.81 (1.86)	*N* = 127 (AG = 127; NAG = 0)	Taekwondo	Training: Junior = 5.1 yrs. (5×/week); Senior = 7.5 yrs. (7×/week); officially registered national athletes; regular competition participation	① State-Trait Anxiety Inventory (STAI) ② Formal Characteristics of Behavior–Temperament Questionnaire (FCZ-KT) ③NEO Personality Inventory-3 (NEO-PI-3)	Grouping: Medalist vs. Non-medalist (Polish Championship); performance: ranking points	Gender distribution not reported; cross-sectional design; Polish athletes only; grouping based on medals without controlling for ranking points, competition level, or training years.
Piepiora (2021)	Poland	0 %	24.5 (NR)	*N* = 300 (CG = 13; NCG	Team sports (soccer, basketball, volleyball, handball, rugby,	≥3 years of professional training, training onset ≥14 years old, with national/	NEO Five-Factor Inventory (NEO-FFI)	International competition medalist (Champion vs. Non-champion)	Limited to Polish male athletes; no female or cross-cultural samples; cross-sectional design; small

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Table 1 (*continued*)

References	Country	Female %	AGE	*N*	Type of sport	Athletic Experience	Personality measurement	Measurement of sports performance	Limitations
				= 287)	ultimate frisbee)	international competition experience.			champion group (*n* = 13); no objective performance data provided (grouping only).
Zar et al. (2022)	Iran	39 %	NR	*N* = 376 (AG = 376; NAG = 0)	Team Sports for Disabled Athletes	All at provincial level or above	Big Five (21-item, Iran)	Categorized by competition level (provincial/ national/ international) via official registration records.	Iranian sample only; training years, sport type, and disability type uncontrolled; personality assessed by self-report; cross-sectional design; proportions of athletes at provincial, national, and international levels not reported.
Palmateer (2022)	United States	0 %	19.46 (1.57)	*N* = 651 (AG = 651; NAG = 0)	Baseball	MLB draft-eligible; completed high school or collegiate baseball competition experience	NEO Personality Inventory-3 (NEO-PI-3)	① Draft order ② Official MLB performance in the first post-draft year (Hitting: BABIP, OBP, SLG; Pitching: BB%, K/BB, ERA)	Player position, coach ratings, and personality changes not considered; short-term follow-up (1 year); small sample (*n* = 23) in low agreeableness group may reduce statistical power; non-drafted athletes not included; limited generalizability as only US MLB draft data were used.
Siemon and Wessels (2023)	Finland & Germany	0 %	NR	*N* = 185 (AG = 185; NAG = 0)	National Basketball Association, NBA	All players had completed at least one full NCAA season, played at least one year in the NBA, and participated in more than 20 NBA games.	① Automated Personality Mining (APM) ② using IBM Watson Personality Insights	① NBA Performance Indicators : PER、WS、WS48、BEPR; ② NCAA Performance Indicators : FG%、FT%、3P %、PPG、RPG、APG、SPG、SOS、WS40	Only players with an active Twitter account were included; personality data may be influenced by the authenticity of social media content; IBM Watson operates as a "black-box" algorithm, lacking transparency; confounding variables such as playing position and cultural background were not analyzed; the study was based on players from 2007 to 2011, limiting the generalizability of the results.
Mi et al. (2023)	中国	39 %	17.37 (NR)	*N* = 139 (AG = 139; NAG = 0)	Basketball	All are officially registered youth basketball players with many years of systematic training experience.	① 16PF (Cattell's Sixteen Personality Factors Test) ② Broad Internal Attention、Broad External Attention ③ Athletic Competition Anxiety Test	Evaluation of athleticism through a combination of coaching evaluations and game performance	Chinese youth sample; gender ratio unspecified; cross-sectional; training years and culture uncontrolled; self-report measures with potential social desirability bias.
Tomczak et al. (2024)	Poland	35 % (AG = 35 %; NAG = 35 %)	20.34 (1.86)	*N* = 289 (AG = 165; NAG = 124)	Mixed sports	performance athletes regularly participated in club training and official competitions	IPIP-BFM-20 (Big Five)	Not directly assessed; classification based on participation level.	Cross-sectional; university students with narrow age range; training years and performance level uncontrolled; self-

(*continued on next page*)

Table 1 (*continued*)

References	Country	Female %	AGE	*N*	Type of sport	Athletic Experience	Personality measurement	Measurement of sports performance	Limitations
						aiming for top achievements.			report measures only, risk of CMV.
Spielmann et al. (2024)	German	27 %	20.43 (4.94)	*N* = 132 (AG = 132; NAG = 0)	Goalkeepers	Registered goalkeepers from 38 German clubs	NEO-FFI (German version)	Grouping: pro, elite youth, semi-pro/ amateur; no performance data	Sample limited to Germany; few female GKs (*n* = 36); grouping bias due to mixed professional and semi-professional players within leagues; no performance data for validation; cross-sectional design.
Kinga and Aleksandra (2024)	Poland	70 % (AG = 80 %; NAG = 50 %)	29.3 (10.71)	*N* = 391 (AG = 159; NAG = 232)	Mixed sports	Actively involved in sporting activities	NEO Five-Factor Inventory (NEO-FFI)	Athlete vs. Non-athlete (proxy for athletic performance)	Cross-sectional; no sport-type or level distinction; training years unreported; gender effect uncontrolled
Gorelová and Halama (2024)	Slovakia	100 %	18.9 (4.08)	*N* = 81 (AG = 81; NAG = 0)	Basketball	Average sports experience: 9.49 years	① Big Five Inventory-2 (BFI-2, Slovak version) ② Sport Emotion Questionnaire (SEQ)	Self-assessment, coach ratings, and Player Efficiency (Slovak Basketball Association)	Small sample (n = 81); female-only; cross-sectional; low reliability in emotion subscales (adjusted by factor analysis); uncontrolled confounders (position, level, training years)
Bonetti et al. (2025)	Brazil & Sweden	25 % (AG = 9 %; NAG = 51 %)	24.6 (5.0)	*N* = 328 (AG = 204; NAG = 124)	Soccer	Elite footballers, at least current players in the Brazilian or Swedish top leagues, some Swedish players with national team experience	Big Five Inventory (BFI)	① Actual performance: number of goals, assists, shots, successful dribbles, etc.; ② Coach ratings (partial results); ③ Classification accuracy of 97 % for elite vs. non-athletes using SVM and ANN.	Cross-sectional design, limiting causal inference; small Swedish sample; limited cultural generalizability as only Brazilian and Swedish players were included; training background and positional differences were not analyzed.

Note. N = Sample size; NR = Not reported; AG = Athlete group; NAG = Non-athlete group; CG = Champion group; NCG = Non-champion group; REG = Regulars; SUB = Substitutes; SUR = Survivors.

and clearer statistical transparency.

6.2. Associations between personality traits and athletic performance

Across the 22 included studies, conscientiousness and extraversion were the most frequently reported personality traits positively associated with athletic performance. Bonetti et al. (2025) found that elite soccer players scored significantly higher than non-athletes on both traits (Conscientiousness: $F(1, 274) = 97.15$, $p < 2.2e\text{-}16$, partial $\eta^2 = 0.26$; Extraversion: $F(1, 274) = 96.12$, $p < 2.2e\text{-}16$, partial $\eta^2 = 0.26$). Similarly, Piepiora (2021) reported higher extraversion (d = 0.55), higher openness (d = 0.70), and lower neuroticism (d = 0.94) among Olympic-level taekwondo athletes compared to controls.

Conscientiousness was consistently identified as a positive predictor across different sport types. It was linked to indicators such as long-term effort, discipline, and goal orientation (Gorelová & Halama, 2024; Piedmont et al., 1999; Ramos-Villagrasa et al., 2013). In contrast, neuroticism showed a negative association with athletic performance. Athletes generally scored lower in neuroticism than non-athletes (Newcombe & Boyle, 1995). Ruiz-Esteban et al. (2020) found that higher neuroticism predicted increased anxiety in female soccer players. Geukes et al. (2013) reported that fear of negative evaluation ($\beta = 0.44$) predicted higher cognitive anxiety, which in turn reduced free-throw accuracy ($\beta = -0.32$). Mediation analysis indicated that anxiety partially mediated this relationship ($R^2 = 0.21$).

Several studies also indicated that high-performing athletes differed from non-elite or amateur athletes in personality profiles. Higher levels of conscientiousness and extraversion, along with lower neuroticism, were repeatedly observed among elite samples (Piepiora, 2021; Remiszewska et al., 2020).

Sport-specific effects were also reported. For example, Monasterio et al. (2018) found gender differences in cooperativeness in a sample of BASE jumpers, though this trait did not show a clear link to performance outcomes.

Table 2
Summary of the Main Findings From the 22 Included Studies on Personality Traits and Athletic Performance.

References	Main findings
Nudel et al. (1989)	① Several athletes scored very high on boldness, warmth, compliance, sensitivity, dominance, drive, self-discipline, and emotional stability; ② The group showed a high average IQ (121); ③ Some athletes had distorted body images linked to anxiety and low self-esteem; ④ Two developed anorexia and one committed suicide; ⑤ Long-distance running did not affect height, growth, or bone development; ⑥ Physiological traits included enlarged left ventricle, high VO_2 max, and few injuries.
Garland and Barry (1990)	① Extraversion, emotional stability, resilience, and group dependence were all significant predictors of performance ($p < 0.05$); ② Athletes'perceived coaching leadership behaviours (training guidance, democracy, rewards, social support) were significantly positively related to performance; ③ Perceived coaching authoritarian behaviours were negatively related to performance; ④ Personality and leadership behaviours interacted to explain 59 % of athletes' performance. The interaction between personality and leadership behaviours explained 59 % of the variance in performance; ⑤ There was a significant triple interaction of Extraversion × Emotional Stability × Resilience, with higher performance being associated with higher levels of these three traits
Newcombe and Boyle (1995)	① Athletes had better overall personality profiles than non-athletes; ② They showed higher extraversion and vigor, and lower anxiety, neuroticism, depression, and confusion; ③ Elite athletes matched Morgan's "Iceberg Profile" with low negative emotions and high vigor; ④ Males showed less anxiety, depression, and neuroticism but greater mental toughness; ⑤ Sport type had little effect on personality, except direct sport athletes had slightly higher vigor.
Egloff and Gruhn (1996)	① Endurance athletes scored significantly higher in extraversion compared to non-athletes ($p < 0.05$); ② Highly trained athletes (>11 h/week) exhibited higher extraversion than moderately trained athletes (<4 h/week) ($p < 0.01$); ③ No significant differences were found between the two groups in neuroticism and openness; ④ Athletes reported significantly fewer physical complaints than non-athletes; ⑤ Neuroticism was associated with the motivation to start exercising for "negative emotion management" and "relaxation"; ⑥ Extraversion was positively correlated with perceived changes in "goal attainment" and "sense of achievement" brought by exercise; ⑦ Training volume was significantly associated with perceived improvements in "physical health."
Piedmont et al. (1999)	① Neuroticism and dutifulness were significantly associated with coaching ratings (negative neuroticism, positive dutifulness); ② dutifulness positively predicted actual game performance ($r = 0.29$, $p < 0.05$); ③ Personality explained 12 % of the variance in coaching ratings and 7 % of the variance in actual game performance
Saale-Prasad (2013)	① Athletes generally exhibit high levels of extraversion, openness, and conscientiousness, and low levels of agreeableness; ② Neuroticism is negatively associated with athletic performance; ③ Concientiousness is a significant predictor of combined academic and athletic performance, and remains significant after controlling for athleticism
Čavala et al. (2013)	① Physical form (body size factor) was not significantly different from competitive level and position; ② Goalkeepers scored significantly lower on extroversion (i.e., introversion-extroversion dimensions) than other positions, showing higher introversion; ③ In the national team alternate group, showed higher ectomorphy (long and lean), whereas the general group showed higher endomorphy (fat component) and mesomorphy (muscle component); ④ athletes did not show significant differences in ectomorphy and neuroticism factors with level of competition
Ramos-Villagrasa et al. (2013)	① Big Five traits, intrinsic motivation, and experience influenced performance differently across the season; ② Conscientiousness consistently predicted performance, especially early in the season; ③ Intrinsic motivation became the strongest predictor at the season's end; ④ Openness was the key factor for performance variability; ⑤ Work experience had limited impact on performance but predicted variability.
Geukes et al. (2017)	① FNE (Fear of Negative Evaluation) was a significant negative predictor of performance in high stress conditions (FNE ↑ → hit rate ↓), and was not valid in low stress conditions; ② Somatic and Cognitive Anxiety was also a significant negative predictor of performance, while Self-Esteem was a positive predictor; ③ Dispositional reinvestment and Athletic identity did not significantly predict performance; ④ state anxiety mediated the effect of FNE on performance, i.e., FNE reduced performance by increasing anxiety; ⑤ performance was differently predicted by real-life competition (high-stress) and laboratory (low-stress) personality.
Monasterio et al. (2018)	① There were no significant differences between males and females on most personality dimensions; ② Females were significantly more likely than males only on the Cooperativeness dimension ($p < 0.05$); ③ BASE Jumpers are generally characterized by high novelty-seeking, low injury-avoidance, high self-directedness, and high co-operation; ④ Gender plays a more important role in High-risk exercise motivation and personality play a lesser role, with self-actualisation and emotion regulation being the main drivers.
Remiszewska et al. (2020)	① Professional athletes scored higher on motivation than amateur athletes ($p = 0.036$); ② Amateur athletes scored higher than professional athletes on stress control ($p = 0.002$) and the impact of performance evaluation ($p = 0.050$); ③ Although professional athletes showed higher psychological skills and team cohesion than amateurs, the differences were not significant; ④ Professional athletes may exhibit lower stress control and resistance to external evaluation due to higher social expectations and competitive pressure; ⑤ Coaches could strengthen stress management and psychological skills training for professional female football players.
Ruiz-Esteban et al. (2020)	① Neuroticism was significantly lower in the Senior group than in the Junior group ($p < 0.001$); ② In the Junior group, medalists were more extroverted than non-medalists ($p = 0.027$); ③ In the Senior group, medalists showed significantly higher agreeableness than non-medalists ($p = 0.030$); ④ Both groups exhibited low neuroticism, high extraversion, and moderate agreeableness.
Piepiora (2021)	① Significant differences in neuroticism, extraversion, agreeableness, and conscientiousness across sports; ② Champion group had lower neuroticism and significantly higher extraversion and openness compared to other athletes ($p < 0.001$); ③ Champion group did not have significantly higher agreeableness and conscientiousness than the average athlete;

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Table 2 (*continued*)

References	Main findings
	④ Personality traits can be used as a focus for psychological selection and training.
Zar et al. (2022)	① Athletes' personality factors (Adaptability, Extraversion, Dutifulness, Neuroticism, etc.) are significantly related to sport performance; ② Extraversion, Adaptability, and Dutifulness are positively related to performance and Neuroticism is negatively related to performance at the national level; ③ Adaptability and Neuroticism are related to performance at the provincial level; ④ Extraversion, Dutifulness, and Neuroticism are significant predictors of performance at the international level; ⑤ Women's Extraversion is more likely to be associated with performance;
Palmateer (2022)	① Based on LPA analysis, players were divided into 2 categories: 96 % were in the high desirability group and 4 % were in the low desirability group; ② There was no significant relationship between draft position and personality category; ③ None of the five major personalities and their facets significantly predicted one-year post-draft performance (Hitting and Pitching); ④ The main predictor variable of post-draft performance was still pre-draft game performance, with personality making a limited contribution; ⑤ Personality Personality is a weak predictor of draft performance, but may be valuable for athlete development and support.
Siemon and Wessels (2023)	① Extraversion ($r = 0.208$–0.273, $p < 0.01$), agreeableness (partially), and conscientiousness ($r = 0.194$–0.242, $p < 0.01$) were significantly positively associated with NBA performance; ② Neuroticism and openness were not significantly associated with NBA performance; ③ Multiple regression analysis showed that NBA performance (PER, WS, BEPR, WS48) could be jointly predicted by extraversion or conscientiousness combined with some NCAA performance indicators.
Mi et al. (2023)	① Chinese youth basketball players generally exhibited high levels of novelty seeking, cooperativeness, emotional stability, adventurousness, mental toughness, self-control, and self-confidence; ② They showed good attention patterns, characterized by strong external attention and focused attention abilities; ③ The average IQ distribution was 121.91, indicating an overall high intellectual level; ④ Youth athletes generally displayed moderately high levels of volitional qualities, such as mental toughness and self-control; ⑤ Male athletes scored higher than females in mental toughness, self-control, decisiveness, and self-confidence ($p < 0.05$); ⑥ A moderate level of anxiety was found to be beneficial for athletic performance.
Tomczak et al. (2024)	① The relationship model was similar for both high-level and recreational athletes; ② Hope fully mediated the link between personality traits (extraversion, conscientiousness, emotional stability, openness) and task orientation; ③ Higher personality traits led to stronger hope and task orientation; ④ Hope did not mediate the link with ego orientation; ⑤ Findings support combining Hope Theory and Achievement Goal Theory to explain athletic motivation.
Spielmann et al. (2024)	① Agreeableness was higher in professional goalkeepers (GK) than in elite youth GKs ($p = 0.009$); ② Male GKs scored significantly lower than female GKs in both agreeableness ($p < 0.001$) and neuroticism ($p = 0.023$); ③ Among female professional GKs, agreeableness was higher than that of all lower-level male GKs; ④ Differences in conscientiousness approached significance in some subgroups ($p = 0.037$) but were not significant after Bonferroni correction; ⑤ No systematic relationship was found between personality traits and performance levels (Elite vs. Pro); the influence of personality on goalkeeping performance was more reflected in individual-environment interactions.
Kinga and Aleksandra (2024)	① Athletes were significantly more extroverted ($p = 0.0434$) and dutiful ($p < 0.0001$) than controls; ② The T allele was more common in athletes ($T = 71$ %, C = 29 %, $p < 0.0001$); ③ There was a significant interaction between DRD2 Taq1D genotype and dutifulness ($p = 0.0004$)
Gorelová and Halama (2024)	① Negative emotions significantly negatively predicted all performance indicators; ② Due diligence was significantly positively related to player self-assessment and coach ratings; ③ Extraversion and openness were significantly positively related to self-assessment and some match statistics; ④ Tension significantly mediated the relationship between extraversion, negative emotions, and performance, with extraversion decreasing tension to increase performance and negative emotions decreasing performance by increasing tension. Emotions decreased performance by increasing tension.
Bonetti et al. (2025)	① Elite players were significantly more extroverted, dutiful, and open than the control group, and significantly less pleasant and neurotic than the control group; ② Elite players showed better working memory, cognitive flexibility, planning ability, and executive functioning; ③ Cognitive ability and personality significantly predicted performance such as scoring, assisting, and succeeding with the ball; ④ The ANN and SVM models can accurately classify elite players based on the ANN and SVM models can accurately classify elite players based on psychological traits with 97 % accuracy

Note. This table summarizes key findings from 22 studies on the link between personality traits and athletic performance, including significant associations, mediators (e.g., motivation, anxiety), and moderators (e.g., sport type, gender). All statistical values are reported as presented in the original studies.

6.3. Mechanisms linking personality traits to athletic performance

Among the 22 included studies, several mechanisms were identified that help explain how personality traits influence athletic performance (Ruiz-Esteban et al., 2020). The most commonly reported were direct effects (9 out of 22 studies), where traits such as conscientiousness, extraversion, and emotional stability were significantly associated with performance outcomes without involving mediators. For example, Bonetti et al. (2025) found that both conscientiousness and extraversion had strong direct effects on match performance indicators like goals and assists (partial $\eta^2 = 0.26$, $p < 0.001$). Similarly, Zar et al. (2022) reported that extraversion ($\beta = 0.231$, $p < 0.01$) and conscientiousness ($\beta = 0.208$, $p < 0.01$) were positively related to competitive classification levels among athletes with disabilities, while neuroticism was negatively correlated with elite status ($r = -0.25$, $p < 0.01$). Saale-Prasad (2013) also demonstrated a direct link between conscientiousness and coach-assessed performance ($\beta = 0.30$, $p < 0.01$), independent of physical attributes.

Moderated effects were identified in at least six studies, indicating that contextual variables influenced the strength or direction of personality-performance associations. For instance, Piepiora (2021) found that conscientiousness was more strongly predictive of success in team sports than in individual sports (interaction effect: $p < 0.05$). Ruiz-Esteban et al. (2020) observed that extraversion was significantly associated with medal-winning outcomes among junior players ($p = 0.027$), whereas agreeableness was more predictive of success among senior players ($p = 0.030$), highlighting age and competition level as potential moderators.

Mediated pathways were also evident. In a study by Gorelová and Halama (2024), emotional stability ($\beta = -0.44$, $p < 0.01$) and extraversion ($\beta = -0.39$, $p < 0.01$) predicted lower levels of competitive

Table 3
Quality assessment–individual evaluation of the studies examined.

References	STROBE Statement—Checklist of items that should be included in reports of cross-sectional studies																					
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Nudel et al. (1989)	YES	YES	Partly	YES	Partly	YES	YES	Partly	NO	YES	Partly	YES	YES	YES	Partly	YES	YES	YES	YES	YES	Partly	YES
Garland and Barry (1990)	YES	YES	YES	YES	YES	YES	Partly	YES	Partly	YES	Partly	YES	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES
Newcombe and Boyle (1995)	YES	YES	Partly	YES	YES	YES	Partly	YES	NO	YES	Partly	YES	YES	Partly	YES	Partly	YES	YES	YES	YES	Partly	YES
Egloff and Gruhn (1996)	YES	YES	Partly	YES	Partly	YES	Partly	YES	NO	YES	Partly	YES	YES	Partly	YES	YES	YES	YES	YES	YES	Partly	YES
Piedmont et al. (1999)	YES	YES	Partly	YES	YES	YES	Partly	YES	NO	Partly	YES	YES	YES	Partly	YES	YES	YES	YES	YES	YES	Partly	YES
Saale-Prasad (2013)	YES	YES	Partly	YES	YES	YES	Partly	YES	NO	Partly	YES	YES	YES	Partly	YES	YES	YES	Partly	YES	YES	Partly	YES
Čavala et al. (2013)	YES	YES	Partly	YES	Partly	YES	YES	YES	Partly	YES	YES	YES	Partly	YES	YES	Partly	YES	YES	YES	YES	Partly	YES
Ramos-Villagrasa et al. (2013)	YES	YES	Partly	YES	YES	YES	Partly	YES	Partly	YES	YES	YES	Partly	YES	YES	YES	YES	Partly	YES	YES	Partly	YES
Geukes et al. (2017)	YES	YES	YES	YES	Partly	YES	YES	YES	YES	YES	Partly	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES	YES
Monasterio et al. (2018)	YES	YES	YES	YES	Partly	YES	YES	YES	Partly	YES	Partly	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES	YES
Remiszewska et al. (2020)	YES	YES	YES	YES	YES	YES	Partly	YES	Partly	YES	YES	YES	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES
Ruiz-Esteban et al. (2020)	YES	YES	YES	YES	YES	YES	Partly	YES	Partly	YES	YES	YES	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES
Piepiora (2021)	YES	YES	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	Partly	YES	YES	YES	YES	YES	YES
Zar et al. (2022)	YES	YES	YES	YES	Partly	YES	YES	Partly	YES	YES	Partly	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES	YES
Palmateer (2022)	YES	YES	YES	YES	Partly	YES	YES	YES	Partly	YES	Partly	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Siemon and Wessels (2023)	YES	YES	Partly	YES	YES	YES	YES	YES	Partly	YES	Partly	YES	YES	Partly	YES	YES	YES	YES	YES	YES	Partly	YES
Mi et al. (2023)	YES	YES	YES	YES	YES	YES	Partly	YES	Partly	YES	YES	YES	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES
Tomczak et al. (2024)	YES	YES	YES	YES	YES	YES	YES	YES	Partly	YES	Partly	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES	YES
Spielmann et al. (2024)	YES	YES	YES	YES	YES	YES	YES	YES	Partly	YES	YES	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES	YES
Kinga and Aleksandra (2024)	YES	YES	Partly	YES	Partly	YES	YES	Partly	NO	NO	Partly	YES	YES	Partly	YES	Partly	YES	YES	YES	YES	Partly	NO
Gorelová and Halama (2024)	YES	YES	YES	YES	Partly	YES	YES	Partly	NO	YES	Partly	YES	YES	YES	Partly	YES	YES	Partly	YES	YES	YES	YES
Bonetti et al. (2025)	YES	YES	YES	YES	YES	YES	YES	YES	Partly	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Note. This table summarizes the quality assessment of 22 cross-sectional studies based on the STROBE checklist. Each column (1–22) corresponds to a specific STROBE item (e.g., title, abstract, background, objectives, methods, results, discussion). "YES" indicates full compliance, "Partly" indicates partial compliance, and "NO" indicates non-compliance.

tension, which in turn enhanced perceived performance efficiency ($\beta = 0.28$, $p < 0.05$), supporting a partial mediation model. Likewise, Geukes et al. (2017) reported that fear of negative evaluation reduced performance ($\beta = -0.22$, $p < 0.05$) through increased state anxiety ($\beta = 0.45$, $p < 0.01$), confirming an indirect effect via cognitive-emotional mechanisms.

Some studies proposed dual-pathway or complex models. For instance, Nudel et al. (1989) found that adolescent long-distance runners with high levels of self-discipline and emotional stability showed elevated VO_2max and beneficial cardiac remodeling. However, the same traits were also observed in cases of psychological maladjustment, including two athletes with anorexia and one suicide, suggesting a potential performance–mental health trade-off.

Other studies reported null associations. Palmateer (2023) found no significant relationships between any of the Big Five traits and one-year post-draft performance among professional baseball players (all $p > 0.05$). Additionally, latent class analysis indicated that 96 % of the sample displayed a high social desirability trait profile, which was not predictive of draft order or subsequent performance.

7. Discussion

This systematic review synthesizes findings from 22 cross-sectional studies and demonstrates that personality traits are significant and contextually modulated predictors of athletic performance. Among the Five-Factor Model dimensions, conscientiousness, extraversion, and low neuroticism consistently emerged as robust indicators of superior performance. These traits were positively associated with various performance metrics, including competition rankings, medal records, and coach evaluations. For instance, conscientiousness—often linked to self-regulation, diligence, and goal persistence—was associated with selection outcomes and stable performance under pressure. Extraversion, characterized by sociability and assertiveness, was beneficial in team-based sports by promoting communication and initiative, while neuroticism showed a consistent negative association with performance, particularly under stress.

Despite these consistent trends, subgroup analyses revealed heterogeneity influenced by sport type, competitive level, and gender. For example, trait–performance associations were typically stronger in team sports than individual disciplines, and elite athletes demonstrated more adaptive personality profiles than their amateur counterparts. Gender-based differences were also observed, with females often scoring higher in neuroticism and agreeableness, traits that may interact with specific psychological demands of their sports. Moreover, openness—linked to creativity and adaptability—demonstrated stronger associations with performance in cognitively demanding or novel environments. These variations reinforce the idea that the utility and expression of traits depend heavily on the psychological and situational demands of the sporting context.

Together, these findings highlight the predictive relevance of stable personality dispositions while emphasizing the importance of considering the interaction between traits and sport-specific environments in athlete development and talent identification.

8. Mechanisms and theoretical frameworks

As illustrated in Fig. 2, personality traits influence athletic performance and athlete selection through three primary pathways: direct neurobiological effects, indirect psychological mediation, and contextual moderation.

Direct effects were the most frequently reported mechanism across the included studies. Traits such as conscientiousness and extraversion showed consistent positive associations with objective performance metrics. For example, Bonetti et al. (2025) demonstrated that elite footballers with high conscientiousness and low neuroticism, assessed using the Big Five Inventory (BFI), achieved significantly superior in-

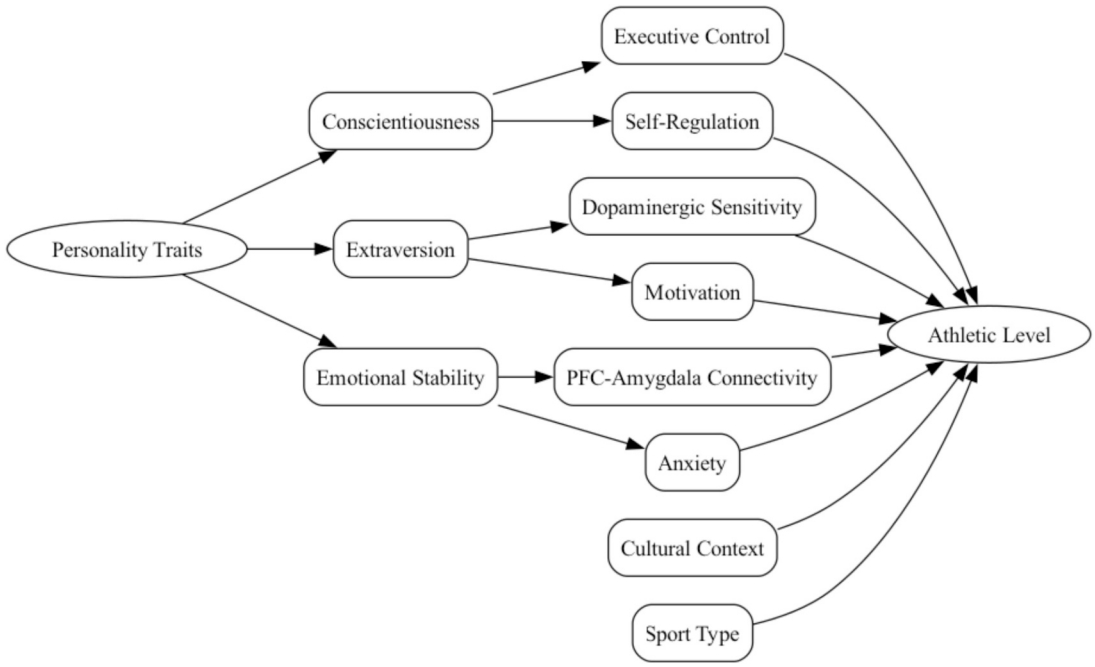

Fig. 2. Mechanism Path Diagram.
*Note.*Tripartite model of mechanisms linking personality traits to athletic performance.

game performance (e.g., goals, assists, successful dribbles). Notably, machine learning analysis based on these traits could classify athletes versus non-athletes with 97 % accuracy. Similarly, Palmateer (2023) found that MLB players scoring higher on conscientiousness and emotional stability (measured by the NEO-PI-3) outperformed their peers on advanced performance statistics such as batting average on balls in play (BABIP), on-base percentage (OBP), and strikeout-to-walk ratio (K/BB). These findings align with neurobiological models suggesting that these traits are underpinned by dopaminergic sensitivity and enhanced prefrontal regulation, which facilitate cognitive stability under stress. Such direct mechanisms have been further substantiated in multiple studies. For instance, Bonetti et al. (2025) and Palmateer (2023) proposed that conscientiousness and emotional stability are closely related to prefrontal cortex self-regulatory functions and dopaminergic system responsiveness. These neurobiological structures play a key role in maintaining cognitive and behavioral stability under pressure, thereby directly enhancing athletic performance.

Indirect effects were observed in studies examining psychological pathways. Ramos-Villagrasa et al. (2013) found that conscientiousness predicted not only composite match ratings but also reduced intra-individual variability across matches, suggesting a stabilizing role over time. Similarly, Roberts et al. (2012) showed that conscientiousness predicted stronger self-regulatory behaviours, which in turn promoted long-term training adherence and performance sustainability. These studies support a mediation model in which personality traits influence motivation and emotional regulation, thereby indirectly shaping performance outcomes.

Moderated effects highlighted the role of sport-specific, demographic, and cultural factors. Piepiora (2021) reported that extraversion and agreeableness were more strongly linked to medal outcomes in team sports than in individual events. Ramos-Villagrasa et al. (2013) further found that football players high in conscientiousness and emotional stability performed more consistently in cognitively and emotionally demanding sports. Moreover, Shuai et al. (2023) documented developmental and cultural shifts, noting that elite endurance athletes demonstrated higher extraversion and emotional stability, possibly due to long-term sport engagement and cultural expectations. These findings emphasize the importance of “trait–task–context” fit in understanding trait utility.

Together, these findings support a tripartite model: (1) direct neurobiological effects, (2) psychological mediation via motivational and emotional processes, and (3) moderated effects influenced by sport type, experience, and cultural environment. This framework suggests that personality traits are not static descriptors but dynamic predictors of readiness and performance. Integrating trait assessment into athlete selection systems may enhance predictive precision, particularly when context-sensitive, developmentally informed, and culturally adapted tools are used. Future research should prioritize longitudinal and cross-cultural studies to refine trait-based evaluation criteria and better delineate the mechanisms underpinning sustained elite performance.

9. Trait variations across sport types and competitive levels

Observed differences in personality traits across sport disciplines and competitive levels offer critical insights into how athlete profiles develop over time. These variations can be interpreted through three theoretical lenses: the Attraction Hypothesis, the Developmental Hypothesis, and the Interactionist Model.

The Attraction Hypothesis posits that individuals gravitate toward sports that align with their innate personality traits. For example, team sport athletes consistently scored higher in extraversion and agreeableness, indicating a preference for social and collaborative environments. However, this model alone cannot account for the personality differences observed across levels of expertise.

The Developmental Hypothesis suggests that sustained involvement in sport can shape personality. Studies have shown that elite endurance athletes display elevated levels of conscientiousness and emotional stability, potentially reflecting psychological adaptations to high-performance demands. Similarly, lower neuroticism among senior Taekwondo athletes suggests emotional regulation may improve through competitive exposure.

Yet, these hypotheses are limited in isolation. The Interactionist Model provides a more integrative perspective, proposing that personality develops through continuous interaction with sport-specific environments. For example, openness may enhance performance in sports requiring adaptive cognition but may be less relevant in routine-based

disciplines. This underscores the importance of aligning individual traits with the demands of specific sports and competitive contexts.

In sum, athlete personality profiles reflect both initial dispositions and ongoing developmental influences shaped by context. Recognizing the reciprocal nature of these relationships can inform more tailored selection strategies and support systems for athletes at different stages of their career.

10. Strengths and limitations

This systematic review provides a structured and comprehensive synthesis of 22 empirical studies examining the relationship between personality traits and athletic performance and selection across diverse sport types, competitive levels, and cultural contexts. In contrast to prior reviews that predominantly focused on the Big Five framework, the present analysis integrates a broader range of personality models—including the Sixteen Personality Factor Questionnaire (16PF), the Temperament and Character Inventory (TCI), and sport-specific psychological constructs—thereby expanding the conceptual scope of personality-performance associations. Moreover, both objective performance indicators (e.g., match statistics, coach ratings) and subjective assessments (e.g., self-reports, psychometric scales) were incorporated, offering a more multidimensional understanding of how personality traits manifest in various athletic contexts.

A further strength of this review lies in the development of a tripartite conceptual framework that systematically organizes the mechanisms linking personality traits to athletic outcomes into three pathways: direct neurobiological effects, indirect psychological mediation, and contextual moderation. This model emphasizes the dynamic, developmental, and context-sensitive nature of personality functioning in sport, providing a theoretical foundation for future applications in talent identification and athlete development.

However, several limitations should be acknowledged. First, most included studies employed cross-sectional designs, limiting causal inference. Second, there was substantial methodological heterogeneity in terms of personality inventories used, performance indicators, analytical approaches, and participant demographics, which constrained comparability and precluded meta-analytic synthesis. Furthermore, many studies lacked adequate control of confounding variables such as gender, sport type, competitive level, training history, and cultural background, all of which may influence the observed associations. Although a comprehensive and systematic search strategy was employed, the possibility of publication bias—particularly the underreporting of null or negative findings—cannot be fully ruled out.

11. Future research directions

To overcome current limitations, future research should adopt longitudinal or experimental designs to clarify causal links between personality traits and athlete selection, especially during early stages of talent development. Incorporating multi-source assessments—such as coach ratings, objective performance metrics, and physiological markers (e.g., heart rate variability, VO_2max)—can enhance measurement validity while minimizing potential biases. Researchers are encouraged to statistically account for key confounders, including gender, training history, cultural background, and sport-specific demands, using multivariate models or structural equation modeling. Standardization of both personality instruments and performance indicators is also necessary to improve study comparability and reproducibility. Lastly, cross-cultural validation studies are essential to examine sociocultural variability in trait-performance relationships and to support the development of culturally responsive athlete selection frameworks.

12. Conclusion

This systematic review synthesizes findings from 22 empirical studies and demonstrates that personality traits—particularly conscientiousness, extraversion, and low neuroticism—are consistently associated with athletic selection and performance outcomes across diverse sports, competition levels, and cultural contexts. These traits showed positive associations with both objective metrics (e.g., match statistics, medal records) and subjective evaluations (e.g., coach ratings, self-assessments), underscoring their predictive value in applied settings.

Three principal mechanisms were identified: direct neurobiological pathways (e.g., trait-related activation patterns), indirect psychological mediators (such as motivation, emotional regulation, and self-discipline), and moderated effects shaped by contextual variables (including sport type, training stage, and cultural background). Although these mechanisms form a coherent conceptual framework, empirical validation remains limited, and further research is needed to substantiate these pathways with robust data.

However, several methodological limitations temper the strength of these conclusions. The predominance of cross-sectional designs, the use of heterogeneous personality assessments, and inadequate control for confounding variables—such as gender, prior experience, and sociocultural factors—restrict causal inference and generalizability. Furthermore, the variability in performance metrics and statistical approaches hinders cross-study comparability.

Future research should adopt longitudinal and experimental designs, implement multimodal personality and performance assessments (e.g., behavioral, psychometric, physiological), and apply rigorous multivariate controls. Standardizing trait measures and performance indicators will be critical for enhancing methodological consistency. Finally, cross-cultural validation studies are essential to refine personality-based selection models and develop culturally responsive talent identification strategies in elite sport.

CRediT authorship contribution statement

Jiaxin Xu: Writing – review & editing, Conceptualization, Data curation, Writing – original draft. **Fengwei Hao:** Conceptualization, Formal analysis, Project administration, Writing – original draft.

Consent for publication

Not applicable. This review does not include individual-level or identifiable personal data.

Ethics approval and consent to participate

This article is a systematic review of publicly available literature and does not involve direct research with human participants. Ethical approval was not required.

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Declaration of competing interest

The authors declare that they have no competing interests.

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Data availability

All data analyzed in this study are extracted from published articles,

which are cited in the manuscript. Full extraction tables and PRISMA checklist are available from the corresponding author upon request.

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